

Xinyu Chen

Ph.D. Candidate

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Google Scholar: <https://scholar.google.com/citations?user=kCewEC0AAAAJ&hl>

Research Interest

Working on the water resources management and sustainability by integrating numerical modeling, multi-source data, and advanced statistical and AI methods

Education

PhD, Civil, Environmental and Sustainable Engineering, Arizona State University, Tempe, USA, ongoing. (Advisor: Enrique R. Vivoni)

Master, Hydrology and Water Resources, Hohai University, Nanjing, China, 2021.

Bachelor, Hydrology and Water Resources, Xi'an University of Technology, Xi'an, China, 2018.

Professional Experience

Aug. 2023 – Present, Graduate Research Associate, in Arizona State University, Tempe, USA

Aug. 2022- Jul. 2023, Research Assistant, in Southern University of Science and Technology, Shenzhen, China

Jul. 2021-Mar. 2022, Engineer, in Changjiang Institute of Survey, Planning, Design, and Research, Wuhan, China

Sep. 2018-Jun. 2021, Graduate Research Assistant, in Hohai University, Nanjing, China

Professional and Academic Engagements

Member, American Geophysical Union (AGU), American Association for the Advancement of Science (AAAS), and American Meteorological Society (AMS).

Reviewer, Environmental Modeling and Software, and Frontiers in Environmental Sciences

Honor & Awards

16th Annual SSEBE Graduate Poster Symposium 1st place award, 2026, ASU

Conference Travel Award, 2025, Arizona State University

AZ Water Scholarship, 2025, AZ Water Association.

15th Annual SSEBE Graduate Poster Symposium 3rd place award, 2025, Arizona State University

Fulton Fellowship, 2023, Arizona State University

Outstanding Master's Thesis of Hohai University, 2022, Hohai University

Grand Prize of the 5th National College Students' Hydraulic Innovation Design Competition, 2017, China Association of Hydraulic Engineering

Professional Skills

Coding and programming languages; GIS and geospatial analysis; Models (hydrologic, climate, statistical, etc.); High-Performance Computing; Big Data Processing and Analysis; Project Management Experience in Engineering

Research Projects

1. Adaptive Hydrologic Forecasting Based on Earth Observations, Hydrologic Modeling, and Advanced Statistical Learning in the Colorado River Basin (*Central Arizona Project Program*), 2026-Present.
2. Managing the Colorado River as an Infrastructure Asset: Fusing Remote Sensing and Numerical Modeling in the Operations of the Central Arizona Project (*NASA Water Resources Applications Program*), 2023-2026.
3. Integrated Space-Airborne-Ground Hydrological Monitoring and Simulation for Watershed Research (*Y01296129*), 2022-2023.
4. The mechanism of runoff generation and concentration in the semi-humid watershed under the strong anthropogenic impacts (*National Key Research and Development Program of China Grants No. 2018YFC1508101*), 2018-2021.
5. The coupling mechanism and simulation of ecological and hydrological process on the heterogeneous underlying surface in the typical humid hilly area (*National Natural Science Foundation of China Grants No. 51879067*), 2018-2020.

Publications

1. **Xinyu Chen**, Liguang Jiang (2025), Emerging patterns of changes in intra-annual variability of streamflow. *Geophysical Research Letters*, 52, e2025GL118724. <https://doi.org/10.1029/2025GL118724>.

2. **Xinyu Chen**, Liguang Jiang, Yuning Luo, and Junguo Liu (2023), A global streamflow indices time series dataset for large-sample hydrological analyses on streamflow regime (until 2022), *Earth System Science Data*, 15(10), 4463-4479. (*As of February 24, 2026, the dataset has been viewed 15,443 times and downloaded 31,245 times from over 60 countries;*
See <https://www.scidb.cn/en/detail?dataSetId=4c6b277bc90f4d5b96d3dccfaf00c80d>)
3. **Xinyu Chen**, Ke Zhang, Yuning Luo, et, al. (2023), A distributed hydrological model for semi-humid watersheds with a thick unsaturated zone under strong anthropogenic impacts: a case study in Haihe River Basin, *Journal of Hydrology*, 129765.
4. **Xinyu Chen**, Quan Quan, Ke Zhang, et al (2021). Spatiotemporal characteristics and attribution of dry/wet conditions in the Weihe River Basin within a typical monsoon transition zone of East Asia over the recent 547 years. *Environmental Modelling & Software*, 143: 105116. doi: 10.1016/j.envsoft.2021.105116 (*ESI Highly Cited Paper*).
5. **Xinyu Chen**, Ke Zhang, Lijun Chao, et al. (2021) Quantifying natural recharge characteristics of shallow aquifers in groundwater overexploitation zone of North China. *Water Science and Engineering*, 14(3): 184-192. doi: 10.1016/j.wse.2021.07.001.
6. Yuhao Wang, Ke Zhang, Yuning Luo, QINUO Zhang, Zhijia Li, Cheng Yao, **Xinyu Chen**, Sheng Wang, Jie Liu, Chuntang Zhang, Shunzhang Li. (2025) GHydroMod-SIM-DAI v1.0: A new distributed hydrological model capable of dynamically recognizing runoff generation mechanisms and accounting for aboveground and underground anthropogenic impacts, *Journal of Hydrology*, Volume 663, Part A, 2025, 134184, <https://doi.org/10.1016/j.jhydrol.2025.134184>.
7. Yuning Luo, Ke Zhang, Wen Wang, **Xinyu Chen**, Jin Feng, Haijun Wang, Wei Liu, Cheng Guo, Cuiying Chen, Xiaozhong Wang, An improved statistical bias correction method for Global Climate Model (GCM) precipitation projection: A case study on the CMCC-CM2-SR5 model projection in China's Huaihe River Basin, *Journal of Hydrology: Regional Studies*, Volume 57, 2025, 102146, ISSN 2214-5818, <https://doi.org/10.1016/j.ejrh.2024.102146>.
8. Yuning Luo, Ke Zhang, Yuhao Wang, Sheng Wang, Nan Wu, Shunzhang Li, QINUO Zhang, **Xinyu Chen**, Hongjun Bao, iRainSnowHydro v1.0: A distributed integrated rainfall-runoff and snowmelt-runoff simulation model for alpine watersheds, *Journal of Hydrology*, Volume 645, Part A, 2024, 132220, ISSN 0022-1694, <https://doi.org/10.1016/j.jhydrol.2024.132220>.
9. QINUO Zhang, Ke Zhang, Lijun Chao, **Xinyu Chen**, et al. A unified runoff

generation scheme for applicability across different hydrometeorological zones. *Environmental Modelling & Software*, 2024, 106138, ISSN 1364-8152, <https://doi.org/10.1016/j.envsoft.2024.106138>.

10. Lijun Chao, Ke Zhang, **Xinyu Chen**, et al (2022). Applicability of WRF-Hydro model based by multi-source precipitation merging in flood forecasting for small and medium-sized watersheds. *Journal of Hohai University (Natural Sciences)*, 50(3): 55-64 (in Chinese). (*Frontrunner 5000 Top Articles in Outstanding S&T Journal of China*)

11. Ke Zhang, Jiaqi Zhou, QINUO Zhang, Xinyu Chen, et al. (2022) Grid-based karst distributed hydrological model. *Water Resources Protection*, 38(01): 43-51 (in Chinese).

12. Ke Zhang, QINUO Zhang, **Xinyu Chen**, et. al. (2021) Gridded Xin'anjiang dual anthropogenic aboveground and underground regulation distributed hydrological model. *Water Resources Protection*, 37(05):94-101+139 (in Chinese).

Presentations

1. “Explaining the Post-2000 Hydrological Shift along with the Millennium Drought in the Colorado River Basin” at AMS26 Annual Meeting, Houston, TX, USA, 2026.
2. “Explaining the Post-2000 Hydrological Shift in the Colorado River Basin” at AGU25 Annual Meeting, New Orleans, LA, USA, 2025.
3. “Uncertainty in Future Hydrology of the Colorado River Basin under CMIP5 and CMIP6” at AGU24 Annual Meeting, Washington, D.C., USA, 2024.

Patents

1. **Chen Xinyu**, Zhang Ke. An a priori parameter estimation method used to estimate the tension water capacity on grid cell scale. *Invention Patent* No. CN 113592154 B.
2. **Chen Xinyu**, Zhang Ke, Zhang QINUO, Huang Pengnian. A runoff generation algorithm based on the dual free water reservoir structure. *Invention Patent* No. CN 113657047 B.
3. Zhang Ke, Luo Yuning, **Chen Xinyu**. A channel flow routing algorithm considering the streamflow regulation of reservoirs. *Invention Patent* No. CN 114840989B.
4. Quan Quan, Wang Yan, Zhou Hao, Wang Yadi, **Chen Xinyu**. A habitat identification algorithm of fishes. *Invention Patent* No. CN 109271694 B.

5. **Chen Xinyu**, Quan Quan, Wang Yadi, Ma Yuming, Zhang Kai, Yang Zichen. A water distiller of low temperature. *Utility Model* No. CN 207551941 U.
6. Su Panzhu, **Chen Xinyu**, Xin Zhenke. A water-saving device used on the close stool. *Utility Model* No. CN 211080456 U.
7. Zhang Siyao, Huang Linmei, Mu Wenxiang, **Chen Xinyu**. A device used to measure water turbidity. *Utility Model* No. CN 208026675 U.

Software Copyright

1. Gridded Xin'anjiang dual anthropogenic aboveground and underground regulation distributed hydrological model (shortened as GXAJ-DAR). *Software Copyright* No. 2021SR1993703.
2. Distributed Xin'anjiang-Haihe model (shortened as DisXAJ-HH). *Software Copyright* No. 2022SR0380897.

Engineering Projects

1. Chongqing Three Gorges Reservoir Area Smart River Chief Construction Project (2021-2022)

I contributed to the top-level architectural design of a digital twin platform for water management, which encompasses the development and upgrade of a suite of application modules spanning hydrological forecasting and early warning, flood and drought disaster management, water resources and ecological monitoring, and hydraulic engineering administration, along with integrated smart water management applications and multi-platform user access portals. In addition, I was responsible for the detailed design of the hydrological and hydrodynamic modeling components within the platform.

2. Water Resource Management System of Ningxia Hui Autonomous Region (2021-2022)

I contributed to the digital twin platform for water resource management, which supports monitoring, forecasting, early warning, simulation, and planning for water quality and quantity using sensor networks and mathematical models.

3. Ministry of Water Resources Small and Medium Rivers Flood Forecasting & Early Warning Platform (2018-2021)

I am responsible for the development of tailored hydrological models for groundwater over-extraction areas in Hebei Province and their integration into national-level system platforms.